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CAMPUS READINESS PLATFORM: A SOFTWARE BASED RESEARCH

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ABSTRACT

Students in engineering colleges who prepare for campus recruitment commonly juggle separate resources: one application for aptitude practice, a different website for coding, and casual guidance from seniors on resumes and interviews. In this paper we introduce the Campus Readiness Platform, a fullstack web application that consolidates these activities into a single system controlled by the institution's Training and Placement (TnP) cell. At its centre is a five-round Placement Drive covering Aptitude,

Technical, Coding, HR, and Verbal rounds, arranged in the same order that major recruiters such as TCS, Infosys, and Wipro follow. Every round is proctored using camera and microphone access along with tab-switch detection; repeated violations trigger forced submission, and a submitted answer can no longer be reopened. Architecturally, the system has three tiers: a React and Vite frontend, a Node.js and Express REST API, and a Neon PostgreSQL database, protected by JSON Web Tokens, bcrypt password hashing, and DOMPurify input sanitisation. The Claude API drives two student-facing features, namely an AI Career Chat that answers placement queries and an AI Mock Interview that runs HR, Technical, or Behavioural sessions through text or voice and ends with a scored report on communication, confidence, and technical depth. In addition, a Domain Explorer describes six career tracks, two independent leaderboards separate complete-drive results from casual quiz attempts, and faculty can upload questions in bulk using a quick form, CSV, or JSON. We built the platform with a design-science approach and assessed it through unit, integration, system, and security testing, in which every executed functional test case passed. During informal trials, ordinary REST calls returned in 200– 500 ms, while AI responses needed 1.5–3 s. Overall, the results indicate that an inexpensive platform owned by the institution can pair sequential proctored testing with AI-based support for placement preparation.

Keywords: Campus Placement Preparation, Proctored Online Assessment, AI-Powered Career Guidance, AI Mock Interview, Examination-Integrity Monitoring, Sequential Assessment Engine, Fullstack Web Application, JWT Authentication, Neon Postgresql, Claude API.

I. INTRODUCTION

For engineering students, the campus placement season carries some of the highest stakes of their college years, but preparation usually happens piecemeal: aptitude practice on one app, coding on another website, resume advice passed on by seniors, and hardly any chance to rehearse a real interview. Institutions such as the School of Engineering and Technology at MGM University operate Training and Placement (TnP) cells that organise recruitment drives, yet these cells seldom have one tool that supports continuous practice. To close this gap, we developed the Campus Readiness Platform, a secure web-based system that gathers the whole preparation process in one place.

The platform mirrors the pattern followed by most Indian campus drives. Rounds on Aptitude, Technical

MCQ, Coding, HR, and Verbal run one after another in the order that recruiters like TCS, Infosys, Wipro, Accenture, and Cognizant use, and an entry on the combined leaderboard is created only once all five rounds are complete. Integrity of the examination was a design goal from the outset. Before a round starts, students

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must permit camera and microphone access and accept a brief set of guidelines; while the test runs, the platform maintains the camera preview, notices when the browser tab is switched, and submits the test automatically after three violations, and every question is locked as soon as the student moves on. Such features resemble the monitoring techniques examined in proctoring research [1], [2].camera and microphone access and accepts a short set of guidelines; during the test the platform keeps the camera preview active, detects browser tab switches, and force-submits the test after three violations, and each question locks once the student moves past it. These features echo the monitoring techniques studied in the proctoring literature [1], [2].

In addition to assessment, the Claude language model supports an AI Career Chat and an AI Mock Interview (HR, Technical, or Behavioural). The interview takes typed or spoken answers and concludes with a structured report that scores communication, confidence, and technical depth. A Domain Explorer presents six career tracks linked to the students' curriculum (IoT, Cyber Security, Blockchain, Software Development, Cloud Computing, and AI/ML), listing salary ranges, skills, certifications, hiring companies, and learning paths. Through a role-based Admin Panel, Faculty, TnP Coordinators, and HODs can maintain the question bank using a quick multi-row form, a CSV upload, or a pasted JSON array.

1.1 Problem Statement

Today, preparing for placements is a fragmented effort. Free practice platforms seldom enforce exam discipline: students may change tabs, search for answers, or repeat a test until the questions are memorised, and as a result many face their first truly proctored assessment without being used to continuous monitoring and a strict one-way question flow. Guidance on careers is similarly inconsistent. Students rarely receive evidence-based advice on the track that suits their strengths, nor an opportunity to practise a realistic interview, and TnP cells have no lightweight tool for managing a large, categorised question bank. A single platform is therefore needed that (a) conducts a sequential placement drive modelled on MNC patterns, (b) enforces exam discipline, (c) offers AI-driven guidance and interview practice, (d) shows career domains with concrete data, and (e) allows faculty to manage questions efficiently, all without licence fees or dependence on a vendor.

1.2 Scope of the Project

The project covers the complete student journey: registration, a personalised dashboard, the five-round proctored Placement Drive, a separate Quick Quiz, two leaderboards, career suggestions based on scores, a resume guide, the AI Career Chat and Mock Interview, and the Domain Explorer, together with a role-based Admin Panel for bulk question upload and student records. We exclude federation across multiple colleges, payment processing, native mobile apps, and integration with recruiters' Applicant Tracking Systems; video-recorded interviews, blockchain certificates, and ML-based placement prediction are left for future work.

II. RELATED WORK

Several parts of placement preparation are already served well. AMCAT offers adaptive computer-based assessments that recruiters trust, HackerRank and LeetCode support coding practice, and company portals provide drills tailored to specific patterns. Academic work on placements has concentrated chiefly on prediction: supervised learning methods such as Naive Bayes and decision trees have been applied to forecast whether students get placed using their academic records [11], and a comparative study assessed multiple machine-learning algorithms for predicting the percentage of students placed [12]. Such studies predict results but do not supply the practice environment itself.

The literature on online exam integrity is substantial too. Atoum et al. [2] merged webcam, microphone, and wearable-camera signals for automated proctoring, while a systematic review of AI-based proctoring systems [1] points out the conflict between exam integrity and privacy, along with ethical concerns. As for interview practice, Naim et al. [5] demonstrated that ratings of interview performance can be predicted from facial expression, language, and prosody; Chou et al. [6] developed an AI mockinterview platform that analyses how candidates perform; and Conversate [7] found, in a study with 19 participants, that follow-up questions and dialogic feedback generated by an LLM make practice feel more realistic. More generally, large language models

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offer both opportunities and challenges for education [8], and recent language-model applications are increasingly used in education and interview practice.

What remains absent is one free system, controlled by the institution, that unites proctored sequential testing, AI-driven career guidance and mock interviews, curriculum-aligned domain exploration, and an efficient admin workflow. Table I sets the proposed platform beside existing free tools and paid vendor products such as AMCAT.

Table I. Comparison of the Campus Readiness Platform with Existing Free and Vendor Systems

Criteria	Existing Free / Vendor Platforms	Campus Readiness Platform
Round Sequencing	Separate quizzes with no connection between them	Genuine five-round sequential drive (Aptitude → Verbal) that follows the actual MNC pattern
Proctoring	Not offered by free platforms	Live camera feed, tab-switch detection, and a one-way answer lock
Leaderboard Design	One combined leaderboard, if present at all	Two separate boards: Placement Drive and Quick Quiz
Career Guidance	Fixed articles, or nothing	AI Career Chat that gives live, context-aware answers
Interview Practice	Absent from comparable free platforms	Complete AI Mock Interview with voice input and output plus scored feedback
Domain Exploration	Kept apart from test practice	Six domains matched to the curriculum, with careers, skills, and roadmaps
Admin Question Management	Entry of one question at a time only	Three ways to upload in bulk: Quick Form, CSV, and JSON
Cost to Institution	Paid licence (e.g. AMCAT)	Free or nearly free hosting; AI usage billed pay-as-you-go at a few dollars
Customization	Question bank fixed by the vendor	Question bank that the institution owns and can edit freely

None of the free, institution-controlled platforms we examined conducts all five recruitment rounds as a single automatic sequential drive in the actual MNC order.

Camera monitoring, tab-switch detection, and one-way answer locking are found in commercial assessment suites [1], [2], but not in the free or college-developed practice tools we examined [9].

Voice-enabled AI mock interviews with scored feedback have been shown in research prototypes [6], [7], yet none is combined with proctored practice tests and career-domain exploration on a platform that a college can control.

College projects of this kind typically add questions to the bank one at a time, and most do not keep full-drive performance apart from casual practice on the leaderboard.

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III. PROPOSED SYSTEM

This section describes the proposed Campus Readiness Platform. It covers the three-tier system architecture, the data flow of the sequential Placement Drive, the database design, and the functional and non-functional requirements the system is built to meet.

System Architecture

The platform follows a three-tier design (Fig. 1) in which presentation, application logic, and data storage are kept apart. The presentation tier is a React single-page application built with Vite; it renders the student portal (Dashboard, Placement Drive, Quick Quiz, Leaderboards, Career Suggestions, Resume Tips, AI Career Chat, AI Mock Interview, and Domain Explorer) as well as the separate Admin Panel, and it communicates with the backend through JSON over HTTPS. The application tier is a stateless Node.js and Express REST API, organised into route modules for authentication, scoring, AI chat, AI interview, and admin operations, with JWT verification guarding the routes that need it; because it keeps no state, it can scale horizontally without a shared session store. Two external services remain independent of one another: a Neon serverless PostgreSQL database that holds persistent data, and the Anthropic Claude API that supplies the Career Chat and Mock Interview feedback.

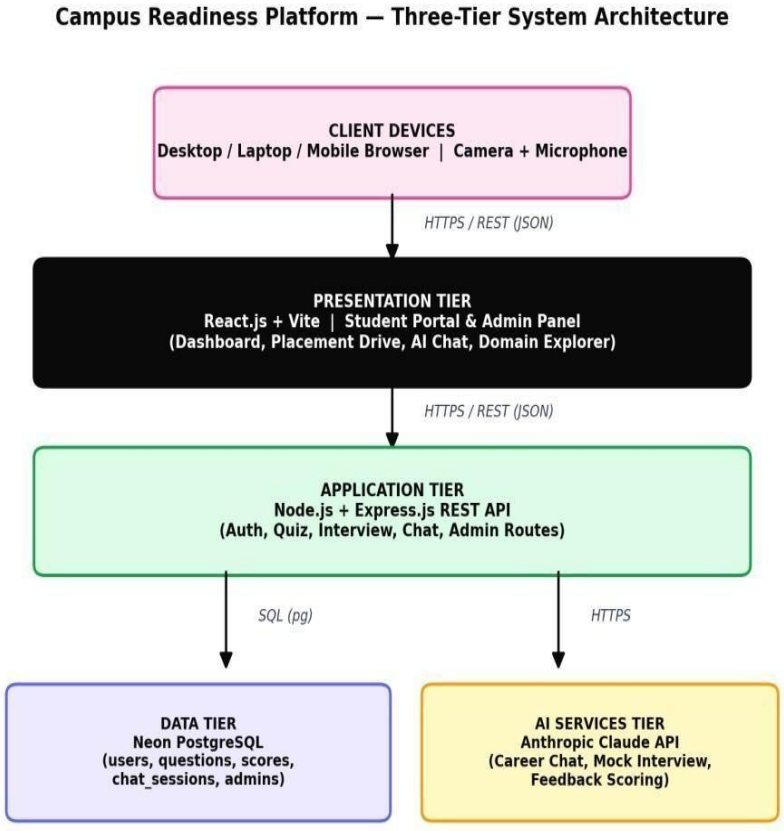


Fig. 1. Three-tier system architecture of the Campus Readiness Platform.

Data Flow

The data flow of the Placement Drive is the most distinctive part of the system, since it has to retain state through five rounds, keep proctoring active at all times, and write a single combined score only when the drive ends (Fig. 2). One shared Proctoring Monitor component serves all rounds, so camera monitoring, tab-switch detection, and the answer lock need to be built and tested just once.

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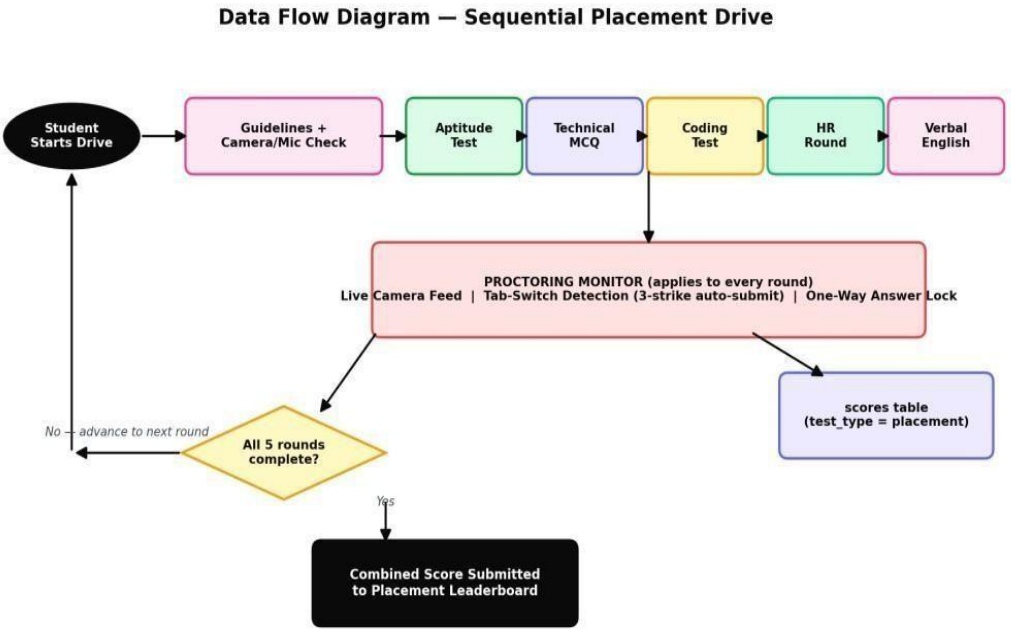


Fig. 2. Data-flow diagram of the sequential Placement Drive.

1.3Database Design

The database schema contains five core tables (Fig. 3). Student and staff authentication are kept separate in the users and admins tables within one PostgreSQL instance; the questions table holds the MCQ bank as flat rows, storing the four options in a text array and tagging each row with a category (aptitude, technical, coding, hr, verbal). The central design decision concerns the scores table: each attempt, whether from the Quick Quiz or a Placement Drive round, is saved with a test_type column ('quiz' or 'placement'), which lets both leaderboards be calculated from one table using a WHERE clause. Finally, a chat_sessions table records Career Chat exchanges so that the daily message limit can be enforced.

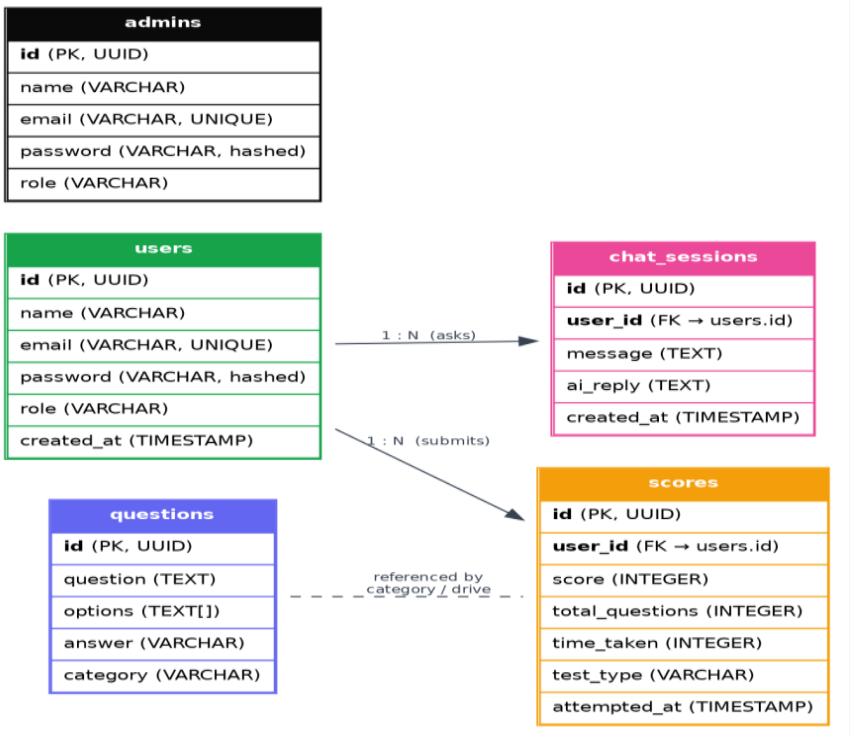


Fig. 3. Entity–relationship diagram of the Campus Readiness Platform database.

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Functional and Non-Functional Requirements

Functional requirements are grouped by module: (1) student registration and login with sanitised input, bcrypt hashing, and JWT sessions; (2) role-based admin authentication using a distinct admin token; (3) a dashboard showing statistics, a placement countdown, and a practice streak; (4) the sequential Placement Drive, where rounds cannot be skipped and the leaderboard entry is written only after every round; (5) a proctored test engine offering a camera preview, three-strike tab-switch auto-submit, and a one-way answer lock; (6) a Quick Quiz of ten questions and ten minutes that stays off the placement leaderboard; (7) two independent leaderboards separated by test_type; (8) career suggestions based on score bands; (9) an AI Career Chat with a daily message limit per student; (10) an AI Mock Interview with optional voice input and output via the Web Speech API, concluding with a structured JSON feedback report; (11) a Domain Explorer covering six career domains; (12) admin question management through quick-form, CSV, and JSON bulk upload that skips invalid rows; and (13) an admin dashboard with searchable student records.

The non-functional requirements are listed below. Performance: API responses within 500–800 ms and initial page content within 2 s. Security: passwords kept only as bcrypt hashes, inputs sanitised using DOMPurify and parameterised queries, JWT expiry checked on each protected route, Helmet.js security headers and restricted CORS, following the OWASP Top Ten. Usability: a responsive interface in which each major module is reachable within two clicks of the dashboard. Reliability: graceful handling when camera or microphone permission is lost, and atomic score submission. Maintainability: backend modules organised by feature and reusable frontend components. Scalability: stateless authentication and a serverless database able to handle an entire class starting a drive simultaneously. Portability: any environment that supports Node.js and a PostgreSQL-compatible connection string, free-tier hosting on Render and Vercel included.

IV. METHODOLOGY

This section explains how the platform was developed and evaluated. It covers the design-science research approach, the data and analytical approach, the tools and technologies chosen, and the fourlevel testing strategy.

Research Design and Development

We followed a design-science approach [10]: the principal contribution is a working artefact that was built and then evaluated against a defined problem and objectives, rather than a hypothesis tested on human participants. Development proceeded iteratively and incrementally, one module at a time, starting from architecture and data-flow documentation, moving through frontend and backend implementation, and ending with added features for assessment, AI feedback, and administration; this allowed us to reconsider architectural decisions without throwing away completed work. Requirements were derived from observing how placement preparation takes place at the institution, within the departmental project-guidance framework, and from the baseline set by industry platforms such as AMCAT.

Data and Analytical Approach

As this is a systems-development project, its “data” consist of technical artefacts, namely architecture and data-flow diagrams together with an ongoing record of build problems and their solutions. The analysis is qualitative and reflective, asking whether the final architecture meets the stated objectives and what engineering trade-offs it required.

Tools and Technologies

We selected the stack for team familiarity, ecosystem maturity, and suitability for the platform’s needs: React with Vite for an interactive assessment interface; Node.js with Express for a REST API written in JavaScript throughout; Neon PostgreSQL for relational integrity across assessment, user, and score data, with managed hosting that enforces SSL; JWT with bcrypt for stateless sessions and one-way password hashing; the Claude API for natural-language interview feedback; DOMPurify to mitigate cross-site scripting; and Render (backend) together with Vercel (frontend) for decoupled deployment.

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Testing Strategy

Testing was carried out at four levels. Unit tests examined individual functions by varying inputs manually, for instance leaderboard ranking with tied and untied scores and resetting the tab-switch counter between attempts. Integration tests spanned the React frontend, the Express API, and the Neon database and concentrated on how Placement Drive state is held across five rounds until one combined score is written; Postman was used to check the request and response contract of every endpoint. In system tests, team members played the role of students and went through a complete five-round drive from registration to the final leaderboard entry, while an admin account carried out bulk uploads. Security tests tried admin-only routes with a student JWT, SQL injection payloads in login and search fields, and raw HTML or script tags in registration and chat inputs.

V. RESULTS AND DISCUSSION

This section reports how the platform performed during testing. It presents the performance evaluation, the test cases and their results, and the observations made during repeated end-to-end drives.

Performance Evaluation

Typical REST operations (retrieving questions, submitting scores, pulling leaderboards) completed in 200–500 ms on Neon's serverless tier under normal load, whereas the AI endpoints needed 1.5–3 s since they rely on an external model call. We repeatedly tested camera and microphone permission, tab-switch detection, and the three-strike auto-submit in Chrome, Edge, and Firefox; in every run the first two violations produced a warning and the third caused the test to submit. Protected routes refused requests without a valid token, student tokens could not access admin endpoints, parameterised queries stopped every SQL injection payload we attempted, and DOMPurify removed script tags from inputs. The structured JSON feedback from the Mock Interview was parsed successfully in most sessions, and a fallback parser handled the rare responses that deviated from the requested schema. Team members and a few volunteer classmates considered the interface intuitive, and they felt the sequential drive reproduced the pressure of a real recruitment test better than a stand-alone quiz.

Test Cases and Results

Table II. Test Cases and Results

ID	Scenario	Input / Action	Expected / Actual Output	Result
TC01	Student Registration & Login	Submit a valid name, email, and password	Account is created, a JWT is issued, and the user lands on the Dashboard	Pass
TC02	Invalid Login Attempt	Use a wrong password for an account that exists	Login is refused with an understandable error message	Pass
TC03	Sequential Placement Drive Order	Begin the drive at the Aptitude round	System moves on automatically through Technical → Coding → HR → Verbal	Pass
TC04	One-Way Answer Lock	Try to go back to a question already answered	Earlier question cannot be opened and appears locked	Pass
TC05	Tab-Switch Auto Submit	Change browser tabs three times while a test is running	Test is submitted automatically at the third violation	Pass
TC06	Camera/Microphone Permission Denial	Refuse camera/mic permission on the Guidelines page	Start button stays disabled and a clear warning is displayed	Pass

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TC07	Dual Leaderboard Separation	Finish a Quick Quiz, then view the Placement leaderboard	Quiz score is absent from the Placement leaderboard	Pass
TC08	Admin Bulk Question Upload (CSV)	Paste a CSV block with several rows and submit it	Every valid row is inserted and a summary of successes is displayed	Pass
TC09	AI Career Chat Response	Put a career-related question to the AI Chat	A relevant AI-generated reply arrives within a few seconds	Pass
TC10	AI Mock Interview Feedback	Finish a mock interview of five questions	A structured feedback report with scores and a recommendation is produced	Pass
TC11	Daily AI Chat Limit Enforcement	Keep sending messages until the daily limit is hit	Additional messages are blocked until the following day	Pass
TC12	Unauthorized Admin Route Access	Call an admin API endpoint with a student token	Request is rejected with a 401/403 status	Pass

Observations

In repeated end-to-end drives using several test accounts at the same time, the platform ran without crashes or data inconsistency, and every score was tied to the right user and test_type, with no duplicate or orphaned rows. Testers said they felt more focused with the one-way answer lock and tab-switch auto-submit than in unproctored quizzes, observed that the two leaderboards gave distinct and sensible rankings, and judged the specific strengths and improvement points in the Mock Interview report to be more actionable than generic interview-tips articles, which agrees with findings that adaptive AI followup questions and feedback make interview practice more realistic [9]. Two limitations emerged: Mock Interview responses were occasionally delayed when the external API had high latency, and voicerecognition accuracy depended on the browser and microphone, working best in Chrome. Neither issue affected core functionality, data integrity, or exam security.

VI. FUTURE SCOPE

At present the platform serves one institution only, and its evaluation was informal and small in scale (team members and volunteer classmates), so we cannot claim better placement outcomes yet. Voice features perform best in Chrome; the AI features rely on the availability of an external API and entail a small cost per use; the Coding round is still multiple-choice instead of a live code editor; and the proctoring layer detects tab switches and the presence of a camera or microphone, but cannot detect faces or multiple people.

Future work includes face and multiple-face detection to strengthen proctoring, an embedded codeexecution engine with hidden test cases for the Coding round, a live admin monitoring dashboard, an ML-based placement-readiness predictor trained on accumulated score history (building on placementprediction research [11], [12]), blockchain-based completion certificates, gamification, a progressive web app, and Hindi and Marathi interface options.

VII. CONCLUSION

This paper described the design, implementation, and testing of the Campus Readiness Platform, a fullstack web application that brings together sequential proctored placement testing, AI-based career guidance and mock interviews, curriculum-aligned domain exploration, and efficient management of the question bank. It is built on React and Vite, Node.js and Express, and Neon PostgreSQL, uses JWT authentication, bcrypt hashing, and input sanitisation throughout, and thus provides a complete threetier architecture. The five-round drive, with camera monitoring, tab-switch detection, and a one-way answer flow, tackles the lack of exam discipline in the free practice tools we examined, while the Claudepowered Career Chat and Mock Interview contribute

structured, scored feedback. Early testing showed the system to be responsive and secure; the main next step is a controlled study of how it affects placement outcomes.

Author Contribution

Each author's contribution followed their responsibilities within the project.

Harshal Kayasth: Responsible for frontend architecture and implementation, covering the React and Vite student portal, the Admin Panel interface, and the UI/UX design of every platform module.

Pawan Solunke: Responsible for backend architecture and development, covering the Node.js and Express REST API, the PostgreSQL database schema, and the JWT-based authentication and security layer.

Vaishnavi Manore: Carried out the literature research and defined the data-collection methodology, and was responsible for research analysis and documentation.

Pratik Mishra: Designed the proctored test engine, handled system and security testing and performance evaluation, and coordinated the project overall.

Monali Maid**:** Provided academic guidance and project supervision, reviewed the system design and implementation, and contributed to the evaluation and refinement of the Campus Readiness Platform. She also supported the preparation and review of the research paper.

Dr. Madhuri Kawarkhe **:** Provided overall research supervision and technical guidance, reviewed the research methodology, results, and analysis, and contributed to the final review and improvement of the research paper.

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Conflict of Interest

The authors declare that they have no conflicts of interest.

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